



传统中药的计算机毒理学研究：简短综述



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摘要
随着存储化合物-靶点/化合物毒性信息的公共数据库以及中药数据库的可用性，计算机模拟方法已被用于中药毒性研究。本文回顾了三种用于毒性研究的计算机模拟方法，包括机器学习、网络毒理学和分子对接。对于每种方法，其应用和实现方式（如单一分类器与多分类器、单一化合物与多化合物、验证与筛选）均进行了探讨。尽管这些方法提供了经过体外和/或体内验证的数据驱动毒性预测，但仍局限于单一化合物分析。此外，这些方法仅限于几种毒性类型，其中肝毒性是最主要的。未来的研究应包括在前端测试化合物组合（即生成用于计算机模拟建模的数据）以及在后端验证预测模型的发现。

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E由 Andreas Bender、Chris de Graaf 和 Noel O’ B 编辑 **oyle**
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关键词
传统中医，网络毒理学，机器学习，分子对接，草药，计算机毒性研究。

介绍
传统中医药（TCM）的使用每年稳步增加。在中国，中医药占

其药品市场份额达到40%，但与西药不同，中医草药通常以多种草药煎剂的形式服用，每种植物的成分协同作用。这本质上会产生多种效果，如抗氧化和抗炎。虽然这一策略体现了更被接受的多靶点药理学模式，但这些草药也不可避免地会产生一些副作用，尽管组合使用可能会中和或抵消这些副作用。与西药不同，中药未受到其疗效和安全性的严格审查，因为它仍被视为补充医学。据估计，每年有约230,000例中医相关的不良反应被报告。此外，草药产品在美国药物性肝损伤的来源中排名第二。尽管为解决这些问题已做出重大努力，

在建立中药草药安全性档案时，可以考虑的一种方法是使用计算机模拟方法。这得益于公共数据库的可用性，这些数据库存储了化合物-靶标/化合物-毒性信息，可用于作用机制和副作用分析。此外，在收集和存储草药植物成分及其化学结构信息到公共数据库方面所做的努力，加快了传统药物化合物分析的进程。一些此类数据库的示例可见于表1，尽管需要注意的是，这个列表并不全面。

在接下来的章节中，将回顾这些数据库在计算机模拟中用于预测和理解中药草药毒性的应用。本小综述涵盖了三种用于中药草药计算机模拟毒性研究的方法，分别是：1) 机器学习，2) 网络毒理学，以及3) 分子对接。需要注意的是，本综述仅涉及化合物的内在毒性，而不包括其外在毒性，即在种植、加工、储存和运输过程中可能导致的草药原料污染。

In silico toxicity studies of traditional Chinese herbal medicine: A mini review

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Abstract
With the availability of public databases that store compound-target/compound-toxicity information, and Traditional Chinese medicine (TCM) databases, *in silico* approaches are used in toxicity studies of TCM herbal medicine. Here, three *in silico* approaches for toxicity studies were reviewed, which include machine learning, network toxicology and molecular docking. For each method, its application and implementation e.g., single classifier vs. multiple classifier, single compound vs. multiple compounds, validation vs. screening, were explored. While these methods provide data-driven toxicity prediction that is validated *in vitro* and/or *in vivo*, it is still limited to single compound analysis. In addition, these methods are limited to several types of toxicity, with hepatotoxicity being the most dominant. Future studies involving the testing of combination of compounds on the front end i.e., to generate data for *in silico* modeling, and back end i.e., validate findings from prediction models will advance the *in silico* toxicity modeling of TCM compounds.

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Keywords
Traditional Chinese medicine, Network toxicology, Machine learning, Molecular docking, Herbal medicine, *In silico* toxicity studies.

Introduction
The use of traditional Chinese medicine (TCM) has steadily increased every year. In China, TCM accounts for

40% of its pharmaceutical market [1] but unlike Western medicine, TCM herbal medicine is usually administered in the form of a decoction of more than one herbal plant, where each constituent of the plant works synergistically. This inherently produces multiple effects such as anti-oxidant and anti-inflammatory. While this strategy exemplifies the more accepted polypharmacology paradigm [2], these herbal medicines also inevitably produce several side effects, although it is possible that the combination of herbs may neutralise or counteract the side effects. Unlike Western medicine, TCM herbal medicines are not subjected to intense scrutiny on their efficacy and safety as it is still considered Complementary Medicine. Every year, it is estimated that 230,000 TCM-related adverse effects were reported [3]. Furthermore, herbal products were ranked second as the source of drug-induced liver injury in the United States [4]. Although significant efforts have been made to address these, the safety profile of TCM herbal medicine is still lacking despite the steady increase in its use.

One approach that can be considered in establishing the safety profile of TCM herbal medicine is the use of *in silico* methods. This is made possible by the availability of public databases that store compound-target/compound-toxicity information that can be used in both mechanisms of action and side effect analyses. Additionally, effort in collecting and storing information on the constituents of the herbal plants and their chemical structure in public databases has expedited the process of analysing traditional medicine compounds. Examples of some of these databases can be seen in [Table 1](#), although it should be noted that this list is not extensive.

In the next sections, the utilisation of these databases *in silico* to predict and understand the toxicity of TCM herbal medicine will be reviewed. There are three different methods covered in this mini review used for *in silico* toxicity studies of TCM herbal medicines, which are: 1) machine learning, 2) network toxicology, and 3) molecular docking. It should be noted that this review only covers the intrinsic toxicity of the compounds but not their extrinsic toxicity, where cultivation, processing, storage and distribution may contaminate the herbal raw material.

Table 1				
List of several databases that are available for <i>in silico</i> toxicity studies of TCM herbal medicine. Note that this list is not exhaustive.				
Database	Description	Notes	URL	Reference
<i>Compound-target database</i>				
ChEMBL	Currently in its 31st version, ChEMBL is a manually curated bioactivity database that contains 2,331,700 compounds and 15,072 targets.	These databases contained bioactivity data points of synthetic and natural compounds. Given the scarcity of bioactivity databases of TCM compounds, ChEMBL and PubChem are useful resources to build machine learning models.	https://www.ebi.ac.uk/chembl	[5]
PubChem	PubChem is an open chemistry database that currently contains 112 million compounds and 185,291 proteins.		https://pubchem.ncbi.nlm.nih.gov	[6]
<i>Protein structure database</i>				
RCSB PDB	The US data center for the global Protein Data Bank (PDB) archive of 3D structure data for proteins, DNA, and RNA molecules	Crystal structures of proteins can be obtained here for docking purposes. However, there is a limited number of membrane proteins available, which may restrict the toxicity study of TCM compounds	https://www.rcsb.org/	[7]
<i>Toxicity-related database</i>				
CTD	CTD (Comparative Toxicogenomics Database) is a manually curated database of chemical–gene/protein interactions, chemical–disease and gene–disease relationships.	These databases provide information on protein targets and their associated toxicity/disease outcome. In network toxicology, this information can be used for screening of toxic TCM compounds and explaining the mechanism of toxicity through further KEGG pathway analysis and GO annotation.	http://ctdbase.org/	[8]
TTD	Therapeutic Target Database (TTD) provides information about the known and explored therapeutic protein and nucleic acid targets, the targeted disease, pathway information and the corresponding drugs directed at each of these targets.		http://db.idrblab.net/ttd/	[9]
OMIM	Online Mendelian Inheritance in Man (OMIM), is a comprehensive, authoritative compendium of human genes and genetic phenotypes. It contains information on all known mendelian disorders and over 16,000 genes.		https://www.omim.org	[10]
<i>TCM-related database</i>				
SymMap	This database contains 698 TCM herbal preparation, 25, 795 ingredients and 20,965 targets of the ingredients	In comparison with other compound databases, the number of herbs contained are small and limited to those registered in the Chinese Pharmacopoeia. Hence, additional data on natural compounds for machine learning models needs to be obtained from other resources e.g ChEMBL	http://www.symmap.org/	[11]
TCMSP	TCMSP (TCM Systems Pharmacology) database contains 499 Chinese herbs registered in the Chinese pharmacopoeia with 29,384 ingredients, 3311 targets and 837 associated diseases.		https://tcmsp-e.com/tcmsp.php	[12]

Reference	
chembl	[5]
ncbi.nlm.nih.gov	[6]
	[7]
	[8]
	[9]
	[10]
	[11]
tcmsp.php	[12]

机器学习模型

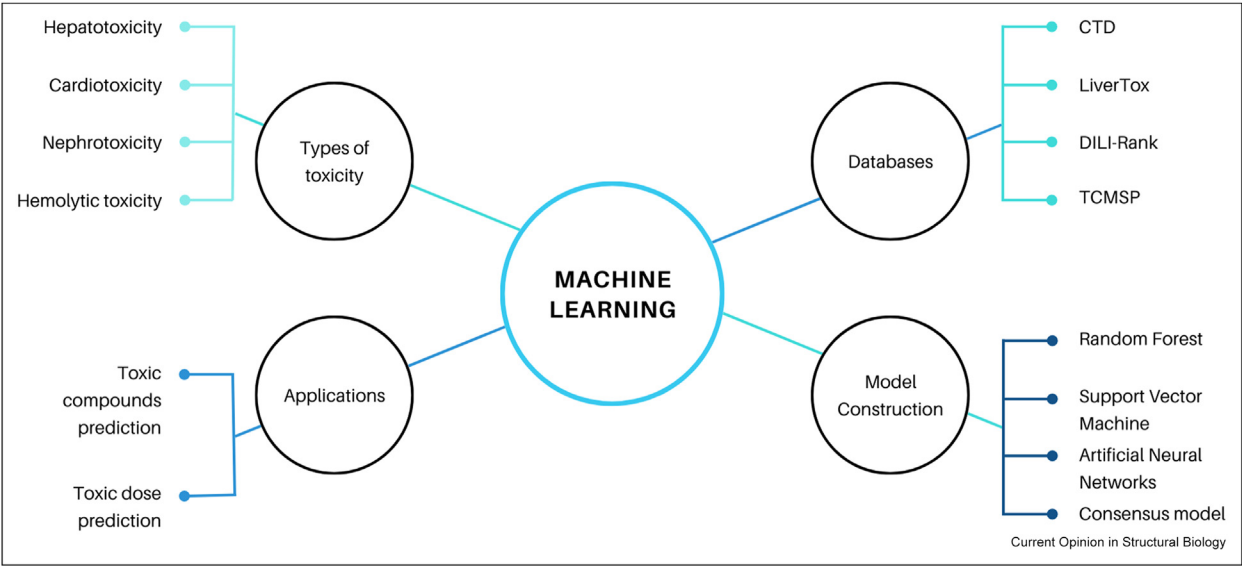
机器学习模型是人工智能领域使用算法模拟人类如何学习解决问题或完成任务。在机器学习模型的训练中，会使用训练集，在毒性预测中，通常使用高质量的数据集，这些数据集包括化合物-靶标或化合物-毒性（见图1）。在前者中，靶标通常是已被证明与特定毒性相关的靶标。关于化合物的信息通常与其结构相关，但也可以包含其理化性质。当仅将化学结构转化为描述符时，会建立基于配体的模型；而当同时转化化学和蛋白质结构时，则会建立蛋白化学计量模型。接着，机器学习算法会将描述符功能性地映射到感兴趣的毒性上。该算法可以基于分类或回归，其中前者预测离散类别标签，而后者

预测 [14e20]。这可能是因为与其他类型的毒性相比，公共数据库中肝毒性数据更为丰富。尽管如此，一些其他类型的毒性也有涵盖，包括心脏毒性 [21]、溶血性毒性 [22]，和肾毒性[23]。这些模型中的大多数是使用诸如 CTD [8]、LivTox [24]，和 DILI-Rank [25]，等数据库的数据构建的，而这些数据库主要包含合成化合物。中药配方仅含天然化合物，并且两类化合物占据不同的化学空间，具有不同的特性。与中药相关的数据库，如 TCMSP [12] 也被使用，但在构建毒性预测模型时，其使用频率并不如非中药特定数据库那样普遍。

大多数建立的中医毒性预测模型是基于分类的，采用的算法包括随机森林（Random Forest） [14,17,20,22]、支持向量机（SVM） [15,16,20] 和人工神经网络（ANN） [17,20]。一些研究采用了多种模型、算法和描述符，以生成更稳健的共识模型来预测中药化合物的毒性 [14,17]。He 等人 [14]使用了朴素贝叶斯（Naive Bayes）、LibSVM、IBK、KStar、AdaboostM1-J48、Bagging-IBK、J48 和随机森林算法来预测何首乌的肝毒性，在外部验证中获得了57.7%至76.3%之间的平衡准确率。随后，这些模型被结合使用朴素贝叶斯分类器生成最终预测结果，组合后的分类器表现更佳，平衡准确率达到78.2%。Wu 等人 [17]，也观察到了类似结果。

自2015年以来，关于机器学习在中医毒性预测中的应用的研究中，大多数研究涉及肝毒性

图 1



使用机器学习方法进行中药毒性研究的概述。大多数涉及机器学习的中药毒性预测研究都集中在肝毒性上，这是由于像DILI-Rank这样的数据库的可用性，该数据库存储了肝毒性数据点。

Machine learning models

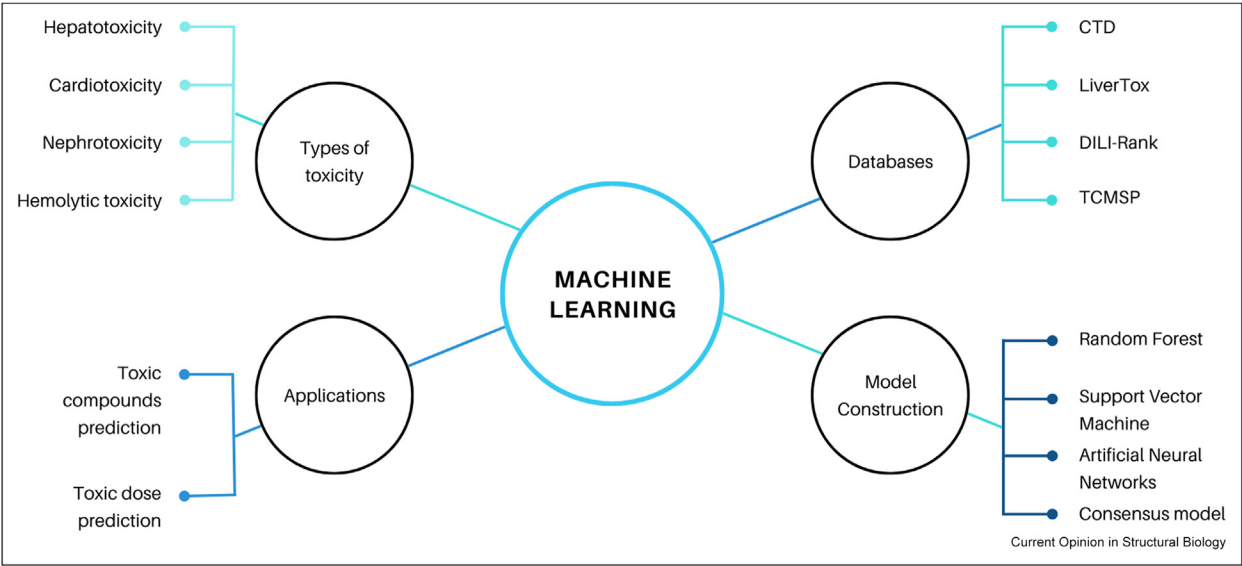
Machine learning is a subfield of artificial intelligence, which employs algorithms to simulate how humans learn to solve a problem or complete a task. A training set is used to train a machine learning model, where in toxicity prediction, high quality dataset consisting of either compound-target or compound-toxicity are usually used (see Figure 1). In the former, the targets are usually those that have been established to be associated with particular toxicity. Information on the compounds is usually related to their structures, but it can also contain their physicochemical properties. A ligand-based model is built when only the chemical structures are transformed into descriptors, and a proteochemometric model is built when both the chemical and protein structures are transformed. A machine learning algorithm will then functionally map the descriptors to the toxicity of interest. The algorithm can either be classification- or regression-based where the former predicts a discrete class label and the latter predicts a continuous value. Examples of machine learning algorithms that are usually used include k-Nearest Neighbour (kNN), Artificial Neural Network (ANN), Random Forest (RF), Naïve Bayes (NB). For more details on the different machine learning available, readers are directed to Ref. [13]. The process then proceeds to validation where models will be subjected to internal and external validations.

When looking at the application of machine learning in predicting TCM toxicities published since 2015, the majority of the studies involved hepatotoxicity

predictions [14–20]. This is likely due to hepatotoxic data being more abundant in public databases compared to other types of toxicity. Nonetheless, several other toxicities have been covered, which include cardiotoxicity [21], hemolytic toxicity [22], and nephrotoxicity [23]. The majority of these models are built using data from databases such as CTD [8], LivTox [24], and DILI-Rank [25], and these databases contain mostly synthetic compounds. TCM formulation contains solely natural compounds, and both types of compounds occupy different chemical spaces and have distinct properties. TCM-related databases such as TCMSP [12] were employed but their use is not as common as non-TCM-specific databases in building toxicity prediction models.

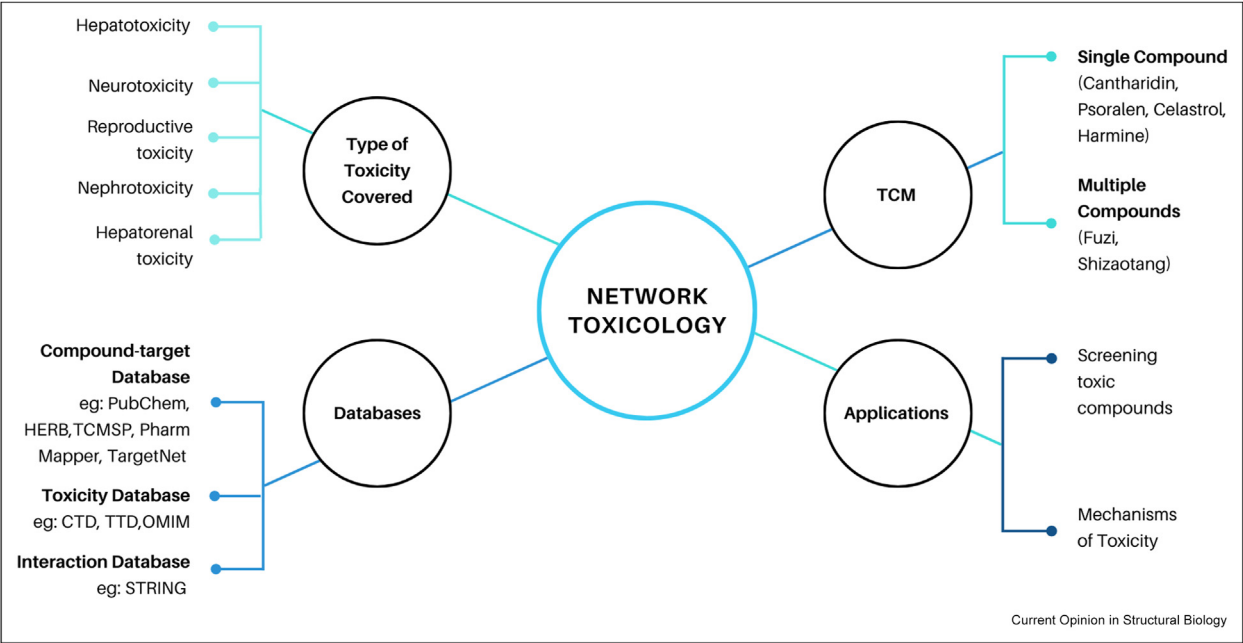
The majority of TCM toxicity prediction models built are classification-based, employing algorithms such as Random Forest [14,17,20,22], SVM [15,16,20] and Artificial Neural Network (ANN) [17,20]. Some studies employed multiple models, algorithms and descriptors to produce more robust consensus models to predict toxicity of TCM compounds [14,17]. He et al. [14], employed NaiveBayes, LibSVM, IBK, KStar, AdaboostM1-J48, Bagging-IBK, J48, and Random Forest algorithms to predict *Polygonum multiflorum* hepatotoxicity resulting in a balanced accuracy between 57.7% and 76.3% in the external validation. Those models were then combined using Naïve Bayes classifier to produce the final prediction where the combined classifier performed better with balanced accuracy of 78.2%. Similar results were observed by Wu et al. [17], when

Figure 1



Overview of TCM toxicity studies using machine learning approaches. The majority of studies involving machine learning in TCM toxicity prediction cover hepatotoxicity due to the availability of databases such as DILI-Rank that stores hepatotoxic data points.

图2



中医药毒理学研究网络概述。利用化合物-靶点、毒性和蛋白质相互作用等各种类型的数据库信息，构建了一个网络，以预测中药配方中的有毒化合物，并建立单一或多种化合物的毒性机制。

使用共识预测模型建立了肝毒性预测模型。共识模型（78%）显示出比最佳单一分类器MACCS-ANN（70%）更高的Matthews相关系数（MCC）。该共识模型被用来筛选TCMSP数据库，结果确定柴胡（*Bupleurum L.*）是含有最多肝毒性化合物的草药。

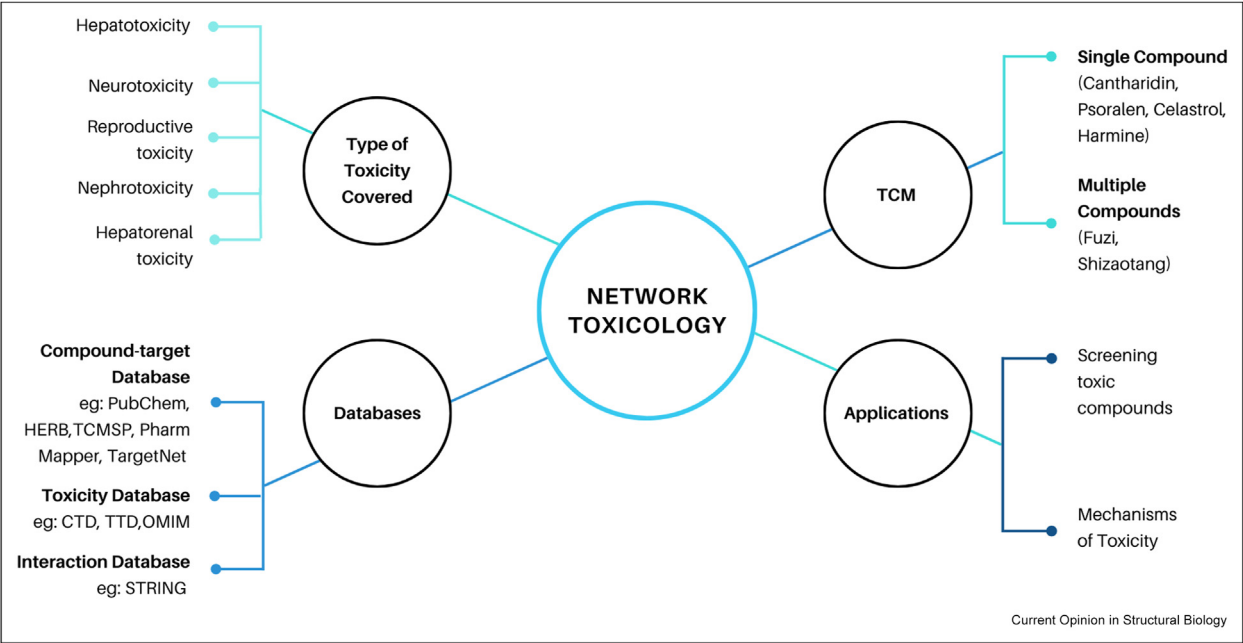
李等人采用不同的方法来进行毒性预测，利用机器学习模型来预测毒性剂量，而不是使用化学结构作为分子描述符。本研究利用体外肝毒性数据，如AUC和实验研究获得的C值。通过逆向建模，预测了与肝毒性概率相对应的剂量方案，针对大黄素、苦参碱、三联啉和氧苦参碱。

尽管深度学习在药物发现中正迅速获得关注，但其在中药毒性研究中的应用仍然有限。迄今为止，仅有一个深度学习模型的相关报道。陈等人 [20],比较了一种名为深度置信网络（DBN）的深度学习模型在筛选潜在肝毒性中西药组合中的表现。结果发现，DBN的分类准确率（CA）为82.2%，优于其他九种机器学习算法，其分类准确率为67.5%至76.7%不等。

网络毒理学
网络毒理学是另一种用于预测中药化合物毒性的计算机模拟方法。网络毒理学研究药物或药物组合如何系统地扰动多个靶点，而不仅仅是引起特定毒性的单个靶点（见图2） [26]。这种方法与中医的整体哲学类似，中草药中多种活性植物化学成分的协同作用作用于多个靶点/通路 [27]。

在构建中药化合物的网络毒理学时，首先为化合物构建化合物-靶点节点，这可以通过从公共数据库如PubChem、HERB、TCMSP、BATMAN-TCM中检索获得，或者通过靶点预测（如Pharm Mapper、Swiss Target Prediction、TargetNet、DRAR-CPI）完成。然后，通过从毒性数据库如CTD、TTD或OMIM中检索信息，构建靶点-毒性节点（见表1）。采用KEGG通路富集分析和基因本体（GO）注释来描述关键基因的特征。接下来，通过系统发育分析和蛋白-蛋白相互作用（PPI）等方法（如使用STRING）建立蛋白节点之间的相互作用。然后，利用拓扑分析等方法构建连接所有可能节点的网络，这是最常用的方法。

Figure 2



Overview of TCM toxicity studies network toxicology. Information from various types of databases such as compound-target, toxicity and protein–protein interaction databases were used to construct a network to predict the toxic compounds in a TCM formulation and establish mechanisms of toxicity of either single or multiple compounds.

consensus prediction model was used to build a predictive model for hepatotoxicity. The consensus model (78%) showed higher Matthews Correlation Coefficient (MCC) than the best single classifier, MACCS-ANN (70%). The consensus model was used to screen the TCMSP database, resulting in the identification of *Bupleurum L. (Chaihu)* as the herb containing the highest number of hepatotoxic compounds.

Approaching toxicity prediction differently, Li et al. [15], utilised machine learning models to predict toxicity dose instead. Rather than using chemical structures as molecular descriptors, this study utilised *in vitro* hepatotoxicity data such as AUC and C_{max} obtained from experimental studies. Through reverse modeling, dosing schedules corresponding to probability of hepatotoxicity were predicted for emodin, matrine, tripolidine and oxymatrine [15].

Although deep learning is fast gaining momentum in drug discovery, its application in TCM toxicity studies is still limited. To date, only one deep learning model has been reported. Chen et al. [20], compared the performance of a deep learning model called deep belief network (DBN) in screening potential hepatotoxic TCM-Western medicine combinations. It was found that DBN outperformed nine other machine learning algorithms with classification accuracy (CA) of 82.2% compared to the rest with a CA of 67.5–76.7%.

Network toxicology
Network toxicology is another *in silico* method used to predict toxicity of TCM compounds. Network toxicology examines how a drug or combination of drugs systematically perturbs multiple targets rather than single targets that cause particular toxicity (see Figure 2) [26]. This approach shares a similar holistic philosophy to TCM, where the synergistic action of multiple active phytochemicals of the herbs act on multiple targets/pathways [27].

In constructing network toxicology for TCM compounds, compound-target nodes are first constructed for a compound by either retrieving from public databases such as PubChem [6], HERB [28], TCMSP [12], BATMAN-TCM [29] and/or subjected to target prediction (e.g. Pharm Mapper [30], Swiss Target Prediction [31], TargetNet [32], DRAR-CPI [33]) The target-toxicity nodes are then constructed by retrieving the information from toxicity databases such as CTD [8], TTD [9] or OMIM [10] (see Table 1). KEGG pathway enrichment analysis and Gene Ontology (GO) annotation are employed to describe the characteristics of hub genes. Next, interactions between the protein nodes are established through methods such as phylogenetic profiling [34] and protein–protein interaction (PPI) [35] using STRING [36]. A network connecting all possible nodes is then constructed using methods such as topological analysis, which is the most often

已采用。拓扑分析通过中心性分析、聚类、最短路径或模体搜索确定节点和边之间的最佳连接度。然后可以使用诸如 Cytoscape [37]和 Pajek [38]等软件对网络进行可视化。

在过去五年中，用于中药毒性预测的网络毒理学大多涵盖肝毒性、神经毒性、生殖毒性、肾毒性和心脏毒性。此外，网络毒理学研究要么针对单一化合物，要么针对中药方剂中的多种化合物。在单一化合物的情况下，网络毒理学通常在体外或体内实验之后进行。He 等人的研究和 Sun 等人的研究分别在进行网络毒理学分析之前，对斑蝥进行基于 UHPLC-QE/MS 的代谢组学分析，对补骨脂进行基于 DARTS 的蛋白质组学分析。He 等人从代谢组学分析中鉴定出 132 个代谢靶点，并额外预测了 38 个靶点。同时，Sun 等人通过基于 DARTS 的蛋白质组学鉴定出 7 个蛋白质，通过预测得到 106 个蛋白质靶点，并通过文献调查获得 45 个靶点。利用这些蛋白质构建了网络毒理学，如前所述。网络显示，斑蝥导致肾

在An等人进行的一项研究中，网络毒理学首先筛选出附子中诱发神经毒性的三种关键有毒化合物（尿嘧啶、棕榈酸、乌头碱）。这些化合物通过十个核心靶点（ALB、AKT1、TP53、IL6、TNF、INS、CASP3、IL1B、EGFR 和 BDNF）通过多种途径诱发神经毒性，如多种神经退行性疾病途径、MAPK 信号通路、HIF-1 信号通路、TNF 信号通路以及各种感染和肿瘤特异性通路。随后，进行了分子对接和体外实验，以验证涉及附子中乌头碱的神经毒性机制。Zhang 等人进行了血清药物化学和网络毒理学研究，以筛选附子的一种加工产品——黑顺片（HSP）的潜在有毒成分。研究确定 HSP 中的 22 种成分可通过调节 AKT1、IL2、F2、GSR 和 EGFR 等靶点，通过多条通路（如 Th17 细胞分化、Jak-STAT 信号通路和谷胱甘肽等）诱发肝毒性。

分子对接

分子对接在原子水平上建立小分子化合物与大型生物分子（如蛋白质、DNA、RNA）之间的相互作用（见图 3）。化合物的结构可以从如 ChEMBL [5] 和 PubChem [6], 等数据库中获取，而生物分子的结构则从蛋白质数据银行（PDB） [7] 获取。如果没有可用的晶体结构，可以通过同源建模获得蛋白质的三维结构。分子对接分为两个阶段：构象采样和评分。采样过程在自由能景观描述的构象空间中进行搜索，其中能量通过评分函数近似计算。这里，评分函数应能够将天然结合构象与能量超曲面的全局最小值关联起来。各种分子对接软件，如 AutoDock [48]和 Glide [49,49], 以及网络服务器，如 PatchDock [50]和 ClusPro [51], 都使用不同的方法进行对接。

当分析多种化合物时，首先进行网络毒理学分析，然后再进行实验验证。网络毒理学首先通过查看与化合物相关的靶点或预测的靶点来筛选配方中哪些化合物具有毒性。与特定毒性相关联的化合物将进一步进行蛋白质-蛋白质相互作用（PPI）分析、KEGG 通路富集分析和 GO 注释。

employed. Topological analysis establishes the best degree of connectivity between nodes and edges through centrality analysis, clustering, shortest paths, or motif search. The network can then be visualized using software such as Cytoscape [37] and Pajek [38].

In the last five years, the majority of network toxicology for TCM toxicity prediction covers hepatotoxicity [39,40], neurotoxicity [41,42], reproductive toxicity [43], nephrotoxicity [44], and cardiotoxicity [45]. In addition, network toxicology is either done on a single compound or multiple compounds of a TCM formulation. In the case of single compound, network toxicology is usually carried out after *in vitro* or *in vivo* experiments. Studies by He et al. [44], and Sun et al. [39], performed UHPLC-QE/MS-based metabolomics on Cantharidin and DARTS-based proteomics on Psoralen, respectively prior to network toxicology. He et al. [44], identified 132 metabolic targets from metabolomics analysis and additional 38 predicted targets. Meanwhile, Sun et al. [39], determined seven proteins through DARTS-based proteomics, 106 proteins predicted targets and 45 targets from literature survey. Network toxicology was constructed using these proteins as explained previously. The network showed that Cantharidin caused nephrotoxicity by inhibiting the glycerophospholipid and sphingolipid pathways, phosphatidylethanolamine, phosphatidylcholine, MAPK3, and PLD2 [44]. Meanwhile, Psoralen induced hepatotoxicity by binding to ABL1, and regulated Nrf2, which increased the level of ROS, and affected the expression of mTOR, leading to cell death [39]. Sun et al. [46], Liu et al. [45], and Sun et al. [41], first confirmed the toxicity of Terrestrosin D (TED), Celestrol and Harmine, respectively *in vivo*, followed by network toxicology. Network toxicology showed that the main targets that played an important role in hepatorenal toxicity induced by TED were HSP90AA1, CNR1, DRD2 through multiple pathways (cAMP signaling pathway, pathway in cancer and G-protein coupled receptor signaling pathway) [41]. Moreover, the ten main targets that induced cardiotoxicity by Celastrol were INS, GAPDH, TNF, VEGFA, CASP3, IL10, MMP9 and MAPK8 through pathways such as TNF, NOD-like receptor and Prolactin signaling pathways [45]. In the case of Harmine, 15 targets were predicted for its neurotoxicity where one of them is acetylcholinesterase. Harmine could directly inhibit the activity of acetylcholinesterase which causes excessive accumulation of acetylcholine.

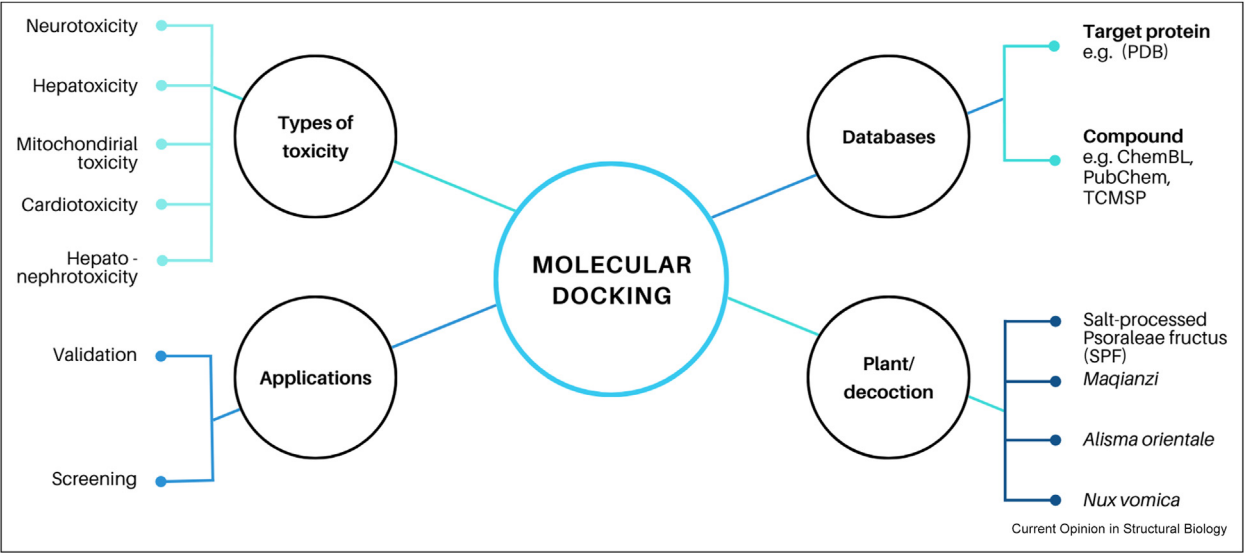
When multiple compounds are being analysed, network toxicology is performed first before experimental validation. Network toxicology first screens which compound(s) in the formulation are toxic by looking at the targets mapped to the compounds or targets predicted. Compounds where the associated targets are linked to specific toxicity will further undergo PPI analysis, KEGG pathway enrichment analysis and GO annotation.

In a study conducted by An et al. [42], network toxicology first screened three key toxic compounds (uracil, palmitic acid, aconitine) in *Fuzi* that induced neurotoxicity. These compounds induced neurotoxicity by ten core targets (ALB, AKT1, TP53, IL6, TNF, INS, CASP3, IL1B, EGFR and BDNF) through pathways such as pathways of neurodegenerative multiple disease, MAPK signaling pathway, HIF-1 signaling pathway, TNF signaling pathway, and various infection and tumor-specific pathways. Then, molecular docking and *in vitro* studies were conducted to validate the mechanism of neurotoxicity involving aconitine in *Fuzi*. Zhang et al. [40], performed serum pharmacochemistry and network toxicology to screen potential toxic components of Heishunpian (HSP), a processed product of *Fuzi*. 22 components of HSP were determined to induce hepatotoxicity by regulating targets such as AKT1, IL2, F2, GSR, and EGFR through multiple pathways (Th17 cell differentiation, Jak-STAT signaling pathway, and glutathione metabolism). In addition, Zhang et al. [47], conducted UFLC-Q-TOF-MS analysis to identify 42 compounds in Shizaotang (SZT) and vinegar Shizaotang (VSZT), followed by network toxicology to look into both effects and toxicity of the formulation. In network toxicology analysis, the top 3 toxic compounds of SZT/VSZT were 3-O-(2,3-dimethylbutanoyl)-13-O-dodecanoyl-20-deoxylingenol, Kansuinin E and (3β)-3-hydroxyanosta-8,24-diene-7,11-dione. Meanwhile, the top 3 targets associated with their toxicity were MAPK3, MAPK1, and AKT1, which affect pathways such as PI3K-Akt signaling pathway, and Epstein–Barr virus infection. Then, these two studies were validated *in vivo*, where the compounds were administered in rats followed by histopathological examination [47].

Molecular docking

Molecular docking establishes the interaction between a small compound and a large biological molecule e.g. protein, DNA, RNA at the atomic level (see Figure 3). The structure of the compound can be obtained from databases such as ChEMBL [5] and PubChem [6], whereas the structure of the biological molecule is retrieved from Protein Data Bank (PDB) [7]. In the event where no crystal structures are available, homology modeling can be used to obtain the three-dimensional structure of proteins. There are two stages in docking, which are conformations sampling and scoring. The sampling process searches the conformational space described by the free energy landscape, where the energy is approximated by the scoring function. Here, the scoring function should be able to associate the native bound-conformation to the global minimum of the energy hypersurface. Various molecular docking software e.g AutoDock [48] and Glide [49,49] and web servers e.g. PatchDock [50] and ClusPro [51] that use different algorithms have been developed to perform the molecular docking exercise. The binding conformation of the protein-ligand complex can be

图3



利用分子对接进行中医药毒理学研究的概述。有几种类型的研究采用对接方法，主要作为网络毒理学、体外和体内研究结果的验证工具。

使用如 PyMOL [52]之类的 3D 可视化软件进行可视化。

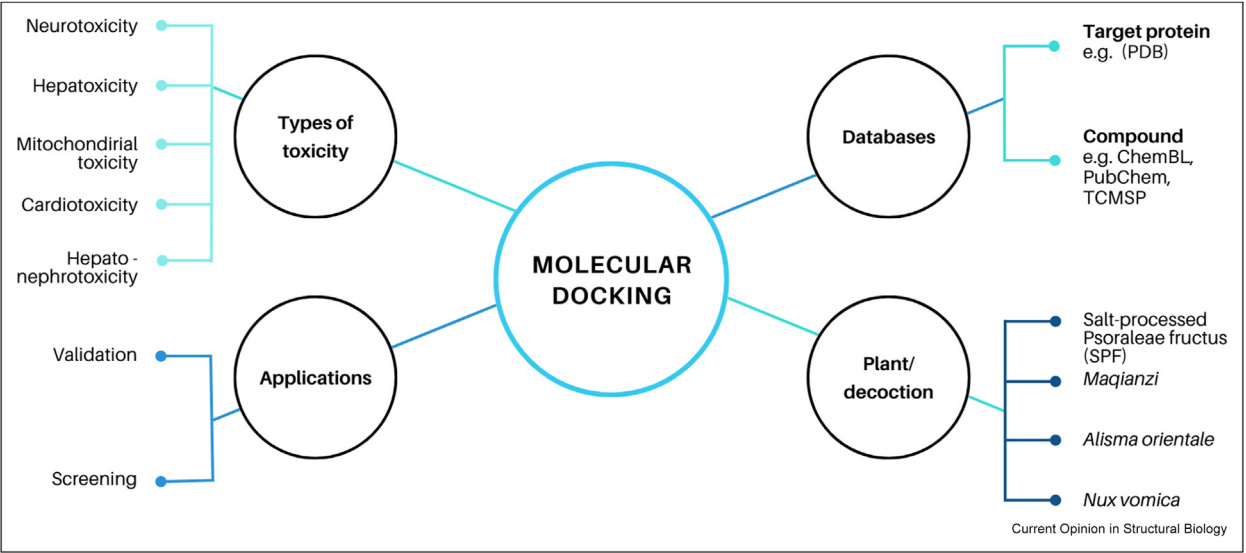
根据过去五年发表的文献，分子对接已被用于预测中药化合物的不同类型毒性，例如肝毒性 [53,54]、神经毒性 [55]，、肝肾毒性 [56]，、线粒体毒性 [57] 以及心脏毒性 [58]。与机器学习不同，对接应用涵盖更多类型的毒性，因为它不依赖于大量的生物活性数据。然而，其应用依赖于已知且经过验证能够引起特定毒性的蛋白靶标的识别。类似于机器学习，对接聚焦于单个化合物，而非像网络毒理学中那样的组合效应。对接对天然产物（例如中药化合物）尤其有优势，特别是在对接浅结合位点时。与小分子相比，自然产物或大分子在对接时的可信度更高。

单独使用分子对接很少用于毒性预测。通常，分子对接会与其他计算方法结合使用，经常用于验证，有时也用于筛选。分子对接通常用于验证网络毒理学 [55,56]，体外 [54] 或体内 [58] 研究的发现。在网络毒理学中，分子对接用于验证网络中的化合物-靶点节点。较低的结合能可以进一步验证潜在的化合物-靶点相互作用，从而缩小范围并识别最可能参与毒性事件的通路和生物过程。Dai 等人。

[55],利用此实验方案来确定马钱子草引起的士的宁神经毒性途径。通过网络毒理学首先识别了蛋白靶点和中枢神经系统相关的基因集合。随后，选择了三种神经毒性化合物（(S)-葡萄苦豆碱、异布鲁定和谷甾醇）及 CHRM1进行对接验证。预测的配体与蛋白CHRM1的结合能低于共晶配体，并形成共晶配体中未见的氢键相互作用，从而确认它们参与了神经毒性途径。张等人 [54]，进行了体外HepG2细胞毒性实验，以及小鼠肝脏中 CYP1A2的基因表达和蛋白印迹分析，其中该酶的活化可加重肝毒性。这些分析旨在测试补骨脂果（PF）对肝脏的作用。进一步进行了对接实验，以确认芳烃受体（AhR）与补骨脂素和异补骨脂素的相互作用。

只有少数研究使用分子对接进行筛选，以识别可能导致目标毒性的化合物。Shen 等人建立了一个整合双重体外检测和分子对接的平台，用于预测 35 种中药注射液的线粒体毒性物质。在初步体外筛选中，共鉴定出 622 种成分为线粒体毒性物质，并进行了分子对接以筛选针对线粒体 NADH 脱氢酶（泛醌）黄素蛋白 1（NDUFV1）的潜在有毒化合物。分子对接结果将潜在毒物缩小到 87 种化合物，其中包括黄芩苷、芸香苷、连香苷和黄芩苷。

Figure 3



Overview of TCM toxicity studies using molecular docking. There are several types of studies that apply docking, where it is mainly used as a validation for results produced by network toxicology, *in vitro* and *in vivo* studies.

visualized using 3D visualizer software such as PyMOL [52].

Based on literature published within the past 5 years, molecular docking has been used to predict different types of toxicity of TCM compounds such as hepatotoxicity [53,54], neurotoxicity [55], hepatonephrotoxicity [56], mitochondrial toxicity [57] and cardiotoxicity [58]. Unlike machine learning, the application of docking covers more types of toxicity as it is not dependent on a substantial amount of bioactivity data. However, its application is dependent on the identification of a protein target(s) that is known and validated to cause certain toxicity. Similar to machine learning, docking focuses on a single compound, and not in combination as in network toxicology. Docking is advantageous to natural products e.g TCM compounds especially when docked against a shallow binding site. Here, the confidence level of natural products or large size molecules docked is higher when compared to small molecules.

Docking on its own is rarely employed for toxicity prediction. In combination with other *in silico* methods, docking is often used for validation and at times for screening purposes. Docking is usually performed to validate findings from network toxicology [55,56], *in vitro* [54] or *in vivo* [58] studies. In network toxicology, docking validates the compound-target node(s) of a network. A lower binding energy would further validate the potential compound-target interaction, which then narrows down and identifies the most likely pathway and biological process that is involved in the toxicity event. Dai et al.

[55], utilised such experimental protocol in determining strychnine-induced neurotoxicity pathway of a *Maqianzi* herb. Protein targets and central nervous system-related gene sets were first identified *via* network toxicology. Following this, three neurotoxic compounds ((*S*)-stylopine, isobrucine and stigmasterol) and CHRM1 were selected for the docking validation. The predicted ligands showed lower binding energy with protein CHRM1 than the co-crystallized ligands and formed hydrogen bond interaction not seen in the co-crystallized ligand, which confirmed their involvement in neurotoxicity pathway. Zhang et al. [54], performed *in vitro* HepG2 cytotoxic assay, and gene expression and western blot analysis of CYP1A2 in mice liver where activation of the enzyme can aggravate hepatotoxicity. These analyses were conducted to test *Psoralea Fructus* (PF) effect on the liver. Further docking exercise was performed to confirm aryl hydrocarbon receptor (AhR) interaction with psoralene and isopsoralene that may be involved in the hepatotoxicity mechanism.

Only a few studies used docking for screening purposes [53,57] to identify compounds that may be responsible for the toxicity of interest. Shen et al. [57], established a platform integrating dual *in vitro* assays and molecular docking, to predict mitochondrial toxicants of 35 TCM injections. From initial *in vitro* screening, a total of 622 ingredients were identified as mitotoxicants, and docking was performed to screen potential toxic compounds against NADH dehydrogenase (ubiquinone) flavoprotein 1, mitochondrial (NDUFV1). The docking result narrowed down the potential intoxicants to 87 compounds, with scutellarin, rutin, phillyrin, and baicalin

这些化合物是对该蛋白具有强亲和力的顶级候选分子。进一步的双重体外实验确认了这些化合物作为线粒体毒物的潜力。同样，吴等人 [53], 通过分子对接筛选了盐制补骨脂（SPF）中可能引起肝毒性的成分。随后进行体内试验，使用不同浓度的SPF煎剂，并进行了包括肝损伤和蛋白质印迹分析在内的多项分析。

对接中的三维可视化可以理解配体与蛋白质之间的相互作用，特别是配体-蛋白复合物中涉及的功能基团。因此，可以生成潜在的结构警示，如Yuan等人所报道的 [59]。在本研究中，观察了檳榔碱和马钱子碱（牛膝草的主要成分）对hERG通道的影响，hERG通道通常用于筛选化合物的心脏毒性。对接结果表明，通过引入氧以补偿化合物的N⁺ 电荷，可能是一种降低hERG活性并增加生物碱类化合物药物开发安全范围的有效策略 [59]。此外，生物碱中存在这一功能基团也可作为未来化合物潜在毒性的警示。

结论

在这篇综述中，介绍了三种用于研究中药材毒性的体外计算方法，分别是机器学习、网络毒理学和分子对接。对于每种方法，回顾了其应用和实施方式，例如单一分类器与多分类器、单一化合物与多种化合物、验证与筛选。总体而言，体外计算方法能够提供基于数据的中药配方潜在毒性预测，并已在体外和/或体内得到验证。然而，这类方法仍然仅限于研究若干毒性。此外，大多数研究仍局限于单一化合物，而中药配方通常涉及多种含有多种化合物的植物。为了提高对不同毒性的覆盖率，除了结构信息，还应考虑其他方法，如文本挖掘。在中药化合物和天然产物数据点有限的情况下，合理的做法可能是寻求来自政府机构、新闻和/或

利益冲突声明

作者声明，他们没有已知的竞争性财务利益或个人关系，可能会影响本文所报告工作的公正性。

数据可用性

本文描述的研究未使用任何数据。

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在审查期间发表的特别感兴趣的论文已被标注为：

* 特别感兴趣 ** 非常感兴趣

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本文对机器学习进行了概览，包括其概念、原理和应用。对于不熟悉机器学习的读者可能会感兴趣。

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being the top candidates with strong affinity towards the protein. Further dual *in vitro* assays of these compounds confirmed the potential of mitochondrial intoxicants. Similarly, Wu et al. [53], screened the components of Salt-processed *Psoralea fructus* (SPF) that could cause hepatotoxicity using molecular docking. Further *in vivo* test was carried out, using SPF decoction at different concentrations and followed by several analyses e.g., liver injury and western blot analyses.

The 3D visualization in docking allows for the understanding of the interaction between ligand and protein, specifically the functional groups that are involved in the ligand–protein complex. Consequently, potential structural alerts can be generated as reported by Yuan et al., [59]. In this study, the effect of strychnine and brucine, the main ingredients of *Nux vomica* were observed on hERG channels which are commonly used to screen for compound cardiac toxicity. The docking result suggested that the presence of an oxygen to compensate for the N⁺ charge of the compound, could be a useful strategy for reducing hERG potency and increasing the safety margin of alkaloid-type compounds in drug development [59]. Furthermore, the presence of this functional group in alkaloids could be used as an alert on potential toxicity of future compounds.

Conclusion

In this review, three different *in silico* methods to study the toxicity of TCM herbal medicines were presented, which are machine learning, network toxicology and molecular docking. For each method, its application and implementation e.g., single classifier *vs.* multiple classifier, single compound *vs.* multiple compounds, validation *vs.* screening, were reviewed. Overall, *in silico* methods provide a data-driven prediction of potential toxicity of TCM formulation that has been validated *in vitro* and/or *in vivo*. However, it is still limited to several toxicities. Additionally, most of the studies are still limited to a single compound when TCM formulation involves multiple plants containing multiple compounds. To increase coverage of different toxicities, other methods such as text mining should be considered in addition to structural information. With the limited data points on TCM compounds and natural products, it may be rational to seek data from government agencies, news and/or social media sites and perform as was done during the COVID-19 pandemic. This may help increase the type of toxicity covered, as well as speed up the progress in this field. However, data from these sources need to be validated first. In the future, more studies should be done to analyse and test the combination of compounds on the front end i.e., to generate data for *in silico* modeling and back end i.e., validate findings from prediction models, of the process.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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作者使用体外实验肝毒性数据和支持向量机（SVM）分类方法，估算了中药衍生天然产物的肝毒性剂量，并将结果与目前推荐剂量进行了比较，发现两者接近。

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本研究通过网络毒理学方法研究了一种名为十枣汤（SZT）的中药配方的毒性作用。这是少数研究中中药配方的研究之一。大多数研究都关注于单一化合物或某种植物的多种化合物。

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本研究强调了分子对接与网络药理学相结合的应用。最初, 采用网络药理学方法筛选马钱子的有毒化合物及其对应的蛋白质, 从而确定神经毒性通路。对接实验证实, 当与CHRM1进行对接时, (S)-stylopine的结合亲和力低于共晶配体。

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缩略词列表

TCM: 中医 SVM: 支持向量机 ANN: 人工神经网络 MCC: 马修相关系数 DBN: 深度置信网络 CA: 分类准确率 SGD: 随机梯度下降 kNN: K近邻 DT: 决策树 RF: 随机森林 LR: 逻辑回归 NB: 朴素贝叶斯 HERB: 高通量实验与参考指导数据库 TCMSP: 中药系统药理学 BATMAN-TCM: 中药分子机制生物信息学分析工具 DRAR-CPI: 通过化学-蛋白互作组进行药物再定位与不良反应 CTD: 比较毒理基因组学数据库 TTD: 治疗靶点数据库 OMIM: 人类孟德尔遗传在线数据库 KEGG: 京都基因与基因组百科全书 GO: 基因本体 PPI: 蛋白质-蛋白质相互作用 STRING: 互作基因/蛋白检索工具 DARTS: 药物亲和靶点稳定性 UHPLC-QE/MS: 超高效液相色谱-四极杆

This study looked into the toxicity effects of a TCM formula called Shizaotang (SZT) by using network toxicology. This is one of the few studies that looked into TCM formulation. Majority of the study looked into single compound or multiple compounds of a plant.

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This study highlights the use of molecular docking in combination with network pharmacology. Initially, network pharmacology method was adopted to screen the toxic compounds of Maqianzi and corresponding proteins, which leads to the neurotoxicity pathway identification. Docking exercise confirmed that (S)-stylopine showed binding affinity lower then the co-crystalized ligands, when docked against CHRM1.

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This study established a platform integrating two *in vitro* assays to enhance the prediction of mitochondrial toxicants from numbers of TCM injections. Initial screening identified 622 ingredients from 4 main TCM injections. Docking narrowed down to 87 compounds, which have strong binding affinity with NDUFV1, hence potential mitotoxicants.

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List of abbreviations

TCM:	Traditional Chinese Medicine
SVM:	Support Vector Machine
ANN:	Artificial Neural Network
MCC:	Matthews Correlation Coefficient
DBN:	Deep Belief Network
CA:	Classification Accuracy
SGD:	Stochastic Gradient Decent
kNN:	K-Nearest Neighbor
DT:	Decision Tree
RF:	Random Forest
LR:	Logistic Regression
NB:	Naive Bayes
HERB:	High-Troughput Experiment- And Reference-Guided Database
TCMSP:	Traditional Chinese Medicine Systems Pharmacology
BATMAN-TCM:	Bioinformatics Analysis Tool For Molecular Mechanism Of Traditional
DRAR-CPI:	Drug Repositioning And Adverse Reactions Via The Chemical-Protein Interactome
CTD:	Comparative Toxicogenomics Database
TTD:	Therapeutic Target Database
OMIM:	Online Mendelian Inheritance In Man
KEGG:	Kyoto Encyclopedia Of Genes And Genomes
GO:	Gene Ontology
PPI:	Protein–Protein Interactions
STRING:	Search Tool For The Retrieval Of Interacting Genes/Proteins
DARTS:	Drug Affinity Responsive Target Stability
UHPLC-QE/MS:	Ultra-High Performance Liquid Chromatography-Quadrupole Time-Of-Flight Mass Spectrometry
UFLC-Q-TOF-MS:	Ultra-Fast Liquid Chromatography Quadrupole-Time Of Flight Tandem Mass Spectrometry
MAPK3:	Mitogen-Activated Protein Kinase 3
PLD2:	Phospholipase D2
ABLI:	Albumin 1
Nrf2:	Nuclear Factor Erythroid 2-Related Factor 2
ROS:	Reactive Oxygen Species
mTOR:	Mammalian Target Of Rapamycin
HSP90AA1:	Heat Shock Protein 90a
CNR1:	Cannabinoid Receptor 1
DRD2:	Dopamine D2 Receptor
cAMP:	Cyclic AMP
INS:	Insulin
GAPDH:	Glyceraldehyde 3-Phosphate Dehydrogenase
TNF:	Tumor Necrosis Factor
VEGFA:	Vascular Endothelial Growth Factor A
CASP3:	Cysteine-Aspartic Acid Protease
IL10:	Interleukin 10
MMP9:	Matrix Metalloproteinase 9
MAPK8:	Mitogen-Activated Protein Kinase 8
NOD-like:	Nucleotide-Binding Oligomerisation Domain-Like
ALB:	Albumin
AKT1:	Akt Serine/Threonine Kinase 1
TP53:	Tumor Protein P53
IL6:	Interleukin 6
IL1B:	Interleukin-1 Beta
EGFR:	Epidermal Growth Factor Receptor
BDNF:	Brain-Derived Neurotrophic Factor
IL2:	Interleukin 2
F2:	Coagulation Factor 2
GSR:	Glutathione Reductase
Th17:	T Helper 17 Cells
Jak-STAT:	Janus Kinase-Signal Transducer And Activator Of Transcription
MAPK1:	Mitogen-Activated Protein Kinase 1
PI3K-Akt:	Phosphatidylinositol 3-Kinase (Pi3k)/Protein Kinase B
PDB:	Protein Data Bank
DNA:	Deoxyribonucleic Acid
RNA:	Ribonucleic Acid
CHRM1:	Muscarinic Acetylcholine Receptor M1
AhR:	Aryl Hydrocarbon Receptor
SPF:	Salt-Processed Psoralea Fructus
NDUFV1:	NADH Dehydrogenase (Ubiquinone) Flavoprotein 1, Mitochondrial
CYP1A2:	Cytochrome P450 1A2
hERG:	Human Ether-A-Go-Go Related Gene